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GITHUB -

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A geospatial analysis system has four sensors supplying images. The percentage of images supplied by each sensor and the percentage of images relevant to a query are shown in the following table.

```
images <- data.frame(sensor = 1:4, Supplied = c(0.15, 0.20, 0.25, 0.40), Relevant = c(0.50, 0.60, 0.80,
```

What is the overall percentage of relevant images?

R = Relevant, S = Supplied

$$P(R) = P(R|S_1) \cdot P(S_1) + P(R|S_2) \cdot P(S_2) + P(R|S_3) \cdot P(S_3) + P(R|S_4) \cdot P(S_4)$$

```
prob_rel <- sum(images$Supplied * images$Relevant)
cat("The overall percentage of relevant images is ", prob_rel * 100, "%.", sep = "")
```

```
## The overall percentage of relevant images is 73.5%.
```

6. A fair coin is tossed twice. Let E_1 be the event that both tosses have the same outcome, that is $E_1 = (HH, TT)$. Let E_2 be the event that the first toss is a head, that is, $E_2 = (HH, HT)$. Let E_3 be the event that the second toss is a head, that is, $E_3 = (TH, HH)$. Show that E_1, E_2 , and E_3 are pairwise independent but not mutually independent.

Sample Spaces

$$S = HH, HT, TH, TT$$

each outcome has an equal probability:

$$P(HH) = P(HT) = P(TH) = P(TT) = \frac{1}{4}$$

Events and Their Probabilities

1. $E_1 = (HH, TT)$
 - $P(E_1) = P(HH) + P(TT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
2. $E_2 = (HH, HT)$
 - $P(E_2) = P(HH) + P(HT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
3. $E_3 = (TH, HH)$
 - $P(E_3) = P(TH) + P(HH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

There are 4 possible outcomes when a coin is tossed twice.

```
sample <- c("HH", "TT", "HT", "TH")
```

```
E1 <- c("HH", "TT")
```

```
E2 <- c("HH", "HT")
```

```
E3 <- c("TH", "HH")
```

```
PE1 <- length(E1) / length(sample)
```

```
PE2 <- length(E2) / length(sample)
```

```
PE3 <- length(E3) / length(sample)
```

```
PE1
```

```
## [1] 0.5
```

```
PE2
```

```
## [1] 0.5
```

```
PE3
```

```
## [1] 0.5
```

FIND THE INTERSECTION

Check if E1 and E2 are independent

```
#### OUTCOMES THAT ARE IN BOTH E1 AND E2
```

```
E1E2 <- intersect(E1, E2)
```

```
#### PROBABILITY
```

```
PE1E2 <- length(E1E2) / length(sample)
```

```
PE1xE2 <- PE1 * PE2
```

```
E1E2
```

```
## [1] "HH"
```

```
PE1E2
```

```
## [1] 0.25
```

```
PE1xE2
```

```
## [1] 0.25
```

Check if E1 and E3 are independent

```
#### OUTCOMES THAT ARE IN BOTH E1 AND E3
```

```
E1E3 <- intersect(E1, E3)
```

```
#### PROBABILITY
```

```
PE1E3 <- length(E1E3) / length(sample)
```

```
PE1xE3 <- PE1 * PE2
```

```
E1E3
```

```
## [1] "HH"
```

```
PE1E3
```

```
## [1] 0.25
```

```
PE1xE3
```

```
## [1] 0.25
```

Check if E2 and E3 are independent

```
#### OUTCOMES THAT ARE IN BOTH E2 AND E3
```

```
E2E3 <- intersect(E2, E3)
```

```
#### PROBABILITY
```

```
PE2E3 <- length(E2E3) / length(sample)
```

```
PE2xE3 <- PE1 * PE2
```

```
E2E3
```

```
## [1] "HH"
```

```
PE2E3
```

```
## [1] 0.25
```

```
PE2xE3
```

```
## [1] 0.25
```

OR

if

$$P(A \cap B) = P(A)P(B)$$

, then vents A and B are independent

1. E_1 and E_2

$$E_1 \cap E_2 = \{HH\}$$

$$P(E_1 \cap E_2) = P(HH) = \frac{1}{4}$$

$$P(E_1)P(E_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Since $P(E_1 \cap E_2) = P(E_1)P(E_2)$, E_1 and E_2 are independent.

2. E_1 and E_3

$$E_1 \cap E_3 = \{HH\}$$

$$P(E_1 \cap E_3) = P(HH) = \frac{1}{4}$$

$$P(E_1)P(E_3) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Since $P(E_1 \cap E_3) = P(E_1)P(E_3)$, E_1 and E_3 are independent.

3. E_2 and E_3

$$E_2 \cap E_3 = \{HH\}$$

$$P(E_2 \cap E_3) = P(HH) = \frac{1}{4}$$

$$P(E_2)P(E_3) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Since $P(E_2 \cap E_3) = P(E_2)P(E_3)$, E_2 and E_3 are independent.

Mutual Independence

```
all <- intersect(intersect(E1, E2), E3)
```

```
p_all <- length(all) / length(sample)
```

```
p_product <- PE1 * PE2 * PE3
```

```
all
```

```
## [1] "HH"
```

```
p_all
```

```
## [1] 0.25
```

```
p_product
```

```
## [1] 0.125
```

The Events Are Not Mutually Independent $E_1, E_2, \dots, E_k$ are mutually independent if:

$$P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_k}) = P(E_{i_1})P(E_{i_2}) \dots P(E_{i_k})$$

From the sets:

$$E_1 \cap E_2 \cap E_3 = \{HH\}$$

$$P(E_1 \cap E_2 \cap E_3) = P(HH) = \frac{1}{4}$$

Calculate $P(E_1)P(E_2)P(E_3)$:

$$P(E_1)P(E_2)P(E_3) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

Since $P(E_1 \cap E_2 \cap E_3) \neq P(E_1)P(E_2)P(E_3)$, then the events are not mutually independent.